VOL. I · NO. 45 THURSDAY, 5 NOVEMBER 2026

The Research Radar

A weekly review of new research from the arXiv preprint server

WEEKLY EDITION · 29 OCT – 05 NOV 2026 · 73 PREPRINTS READ · 21 SELECTED

INSIDE THIS EDITION

CS.CL
Computation and Language
3 reports, 6 in brief

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Systems and Control
3 reports, 0 in brief

SECTION · ARXIV CS.CL

Computation and Language

Coverage: conversational AI, dialogue systems, LLM agents and retrieval-augmented generation.

THE WEEK IN BRIEF

This week's research highlights advancements in LLM agents and dialogue systems, focusing on memory frameworks, procedural organization, and multi-turn reinforcement learning. Several papers explore specialized capabilities such as prospective memory for proactive agents, context enrichment, and speech-LLM memory interference. Additionally, studies investigate human-agent collaboration through decision consultation and evaluate dialogue evaluation judges.

LEAD STORY · CS.CL · COMPUTATION AND LANGUAGE

Prospective Memory for Proactive LLM Agents

The paper introduces Pro-Memory, a cognitively inspired memory framework that equips proactive LLM agents with prospective memory to retain future intentions and realize them at the right time.

BY YUANCHEN BEI, ZHINING LIU, TIANXIN WEI AND OTHERS · SUBMITTED 3 NOVEMBER 2026 · ARXIV:2611.05731

THE PROBLEM

Most existing LLM agent memory systems are retrospective—they retrieve past information to inform current decisions—which fails to address the needs of proactive agents that must remember future-directed intentions and decide when to act without explicit user requests.

THE APPROACH. The authors propose Pro-Memory, a framework that separates the maintenance of future intentions from deciding when and how to realize them. It preserves pending objectives with supporting evidence, assesses new observations against activation conditions, updates intention states based on actions or resolution, and uses an experience memory to distill feedback from completed episodes.

KEY FINDINGS

- Experiments across two benchmarks (delayed-intention tasks and open-ended proactive assistance) show consistent improvements in proactive task performance over representative memory approaches.
- The framework operates successfully across multiple LLM backbones.
- It limits unnecessary interventions while timing actions appropriately.

WHY IT MATTERS

"For LLM agents and conversational dialogue systems, transitioning from passive retrospective retrieval to active prospective memory is crucial for enabling proactive assistance, allowing agents to initiate helpful actions at the correct moment without causing disruptive or premature interventions."

LIMITATIONS. The current study focuses primarily on text-based proactive interactions rather than multimodal signals (like images, audio, and video), and does not yet address complex long-horizon scenarios with interdependent intentions or real-world safety guarantees.

Filed under: prospective memory · proactive llm agents · conversational ai

READ THE ABSTRACT | FULL PAPER (PDF) | DISCUSS THIS PAPER

ALSO ONLINE Paper page - Beyond Static Summarization: Proactive Memory E (huggingface.co) · GitHub - TeleAI-UAGI/Awesome-Agent-Memory: Curated systems, (github.com) · Suggestion: Consider Citing "A-Mem: Agentic Memory for LLM A (github.com)

REPORT · CS.LG · MACHINE LEARNING

What Does Multi-Turn Reinforcement Learning Really Improve in LLM Agents?

Multi-turn agentic reinforcement learning improves long-horizon LLM agent performance, but these gains heavily depend on having sufficient interaction turn budgets.

BY KEJIAN ZHU, ZHUORAN JIN, SHANGQING TU AND OTHERS · SUBMITTED 4 NOVEMBER 2026 · ARXIV:2611.06483

THE PROBLEM

It is poorly understood what capabilities multi-turn agentic reinforcement learning (RL) actually improves in LLM agents, and how reliable these agents are under realistic interaction constraints.

THE APPROACH. The authors evaluate agentic RL using Pass@(K,T) metrics across sampling breadth (K) and interaction depth (T), analyze step-level action quality, and introduce BudgetMix RL to vary interaction budgets during training.

KEY FINDINGS

- Trajectory-level RL advantages depend on sufficient turn budgets and diminish significantly when budgets tighten.
- Agentic RL does not uniformly elevate single-step decision quality; its strength concentrates in error recovery and long-horizon resilience.
- BudgetMix RL achieves robust performance across diverse execution settings by varying interaction budgets during training.

LIMITATIONS. The gains of agentic RL shrink under tight interaction budgets, and single-step action quality does not consistently improve, presenting deployment risks in applications with strict budgets or irreversible actions.

WHY IT MATTERS. It demonstrates that LLM agents and conversational systems evaluated solely on final task success may fail under tight interaction constraints, highlighting the need to evaluate robustness across execution constraints and individual decision reliability.

Filed under: agentic rl · interaction budget · llm agents

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ALSO ONLINE GitHub - WooooDyy/AgentGym-RL: Code and implementations for (github.com) · Context-lite Multi-turn Reinforcement Learning for LLM Agent (openreview.net) · Awesome-Multi-Turn-LLMs/README.md at main · yubol-bobo/Aweso (github.com)

REPORT · CS.AI · ARTIFICIAL INTELLIGENCE

WASP-Mem: Let Agent Organize Procedural Memory. Human Spend Effort on Diagnosis

The paper introduces WASP-Mem, a training-free method that consolidates execution traces and feedback into persistent wiki pages to improve LLM agent procedural memory.

BY PEIZHI NIU, GE SHI, VAIBHAV GORDE AND OTHERS · SUBMITTED 2 NOVEMBER 2026 · ARXIV:2611.04003

THE PROBLEM

Large language model (LLM) agents struggle to automate complex enterprise procedures because static harnesses cannot capture proprietary policies, undocumented behaviors, or state-dependent recovery paths that only appear during execution.

THE APPROACH. The authors propose WASP-Mem (Wiki-based Agent-Synthesized Procedural Memory), which distills execution traces and feedback into persistent, interlinked wiki pages to enable continual agent learning without changing model parameters.

KEY FINDINGS

- WASP-Mem raises the checkpoint pass rate by 4.6-6.1% over the strongest baseline across three models while requiring 57-61% less storage.
- Prescriptive knowledge organization, such as human-defined schemas and explicit deduplication rules, hurts performance.
- Human diagnosis converts failed traces into corrective knowledge and drives sustained improvement.
- A single round of human diagnosis outperforms four rounds of self-judgment by 7.0-10.9%.

LIMITATIONS. Not stated in the provided text.

WHY IT MATTERS. It demonstrates an effective, training-free method for long-term procedural memory management in LLM agents, outlining a clear division of labor where agents organize memory and humans focus selectively on failure diagnosis.

Filed under: llm agents · procedural memory · continual learning

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ALSO ONLINE GitHub - Tavish9/awesome-daily-AI-arxiv: 🚀 Daily AI Research (github.com) · GitHub - js-lee-AI/awesome-llm-agent-papers: LLM agents: cur (github.com) · GitHub - PierrunoYT/awesome-ai-dev-tools: A curated list of (github.com)

IN BRIEF · MORE FROM CS.CL

A Framework for Context Enrichment and Selective Robustness in LLM Social Agents

Learning When Human Decisions Matter: Executor-Relative Consultation Value for LLM Agents

Optimize the Harness, Not the Model: Fitting a Minimal Agent to a Self-Hostable LLM for Investigative Analysis

Mattia Greiche, Reihaneh Rabbany and others · cs.MA · 4 November 2026

The Right Source Is Not Enough: Understanding History Interference in Speech-LLM Memory

HongJin Song, Weiqiao Shan and others · cs.CL · 3 November 2026

Shuli Jiang, Khushbu Pahwa and others · cs.CL · 4 November 2026

Do Multi-Turn Conversational Capabilities Carry Over from Text to Speech? A Parallel Multilingual Benchmark

Jiaming Zhou, Pei Zhang and others · cs.CL · 3 November 2026

Shai Nahum Gefen, Vered Chassida and others · cs.AI · 2 November 2026

Same Rule, Different Verdicts: Rubric Language and Clause Structure in LLM Judges for Korean Support Dialogues

Sangdo Han and Moonsoo Kim · cs.CL · 3 November 2026

ELSEWHERE IN THE JOURNALS

Peer-reviewed and web sources from the past month, beyond arXiv.

ACLANTHOLOGY.ORG

AdaMemPIP: Adaptive Memory Pipeline Orchestration for LLM Agents

Yannis Katsis, Sara Rosenthal, Kshitij Fadnis, Chu-laka Gunasekara, Young-Suk Lee, Lucian Popa, Vraj Shah, Huaiyu Zhu, Danish Contractor, and Marina Danilevsky. 2025. mtrag: A...

ACLANTHOLOGY.ORG

A Survey on Long-Term Memory Security in LLM Agents

Retrieval-augmented generation for knowledge-intensive NLP tasks. In Advances in Neural Infor-mation Processing Systems, volume 33, pages 9459– 9474. Curran Associates, Inc....

ACLANTHOLOGY.ORG

Mindscape-Aware Retrieval Augmented Generation for Improved ...

Laurens van der Maaten and Geoffrey Hinton. 2008. Visualizing data using t-sne. Journal of machine learning research, 9(Nov):2579–2605. Adyasha Maharana, Dong-Ho Lee, Sergey...

ACLANTHOLOGY.ORG

AgenticLOCI: A Multi-Agent Memory Framework for Conversational ...

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Hein-rich Küttler, Mike Lewis, Wen-tau Yih, Tim Rock-täschel, Sebastian Riedel, and...

SECTION · ARXIV CS.LG

Machine Learning

Also drawing on cs.AI, cs.CV, cs.CL, cs.DC. Coverage: energy-efficient and sustainable machine learning: lower compute, carbon and cost.

THE WEEK IN BRIEF

This week's research focuses on energy-efficient and sustainable machine learning, with a strong emphasis on large language models and edge computing. Multiple papers explore reducing the energy consumption, compute costs, and hardware demands of deploying language models across diverse domains like medical question answering and security operations. Additionally, studies investigate specific optimization strategies, including dynamic GPU frequency control, low-bit decoding, and efficient representations for resource-constrained devices.

LEAD STORY · CS.DC · DISTRIBUTED, PARALLEL, AND CLUSTER COMPUTING

Energy-Efficient Reinforcement Learning for Large Language Models

EE-RL is an online controller that dynamically selects separate GPU clock frequencies for the generation and training phases of large language model reinforcement learning, reducing energy consumption while preserving throughput.

BY RAHUL SHARMA, NITIN VEGESNA AND YANXIN ZHANG · SUBMITTED 3 NOVEMBER 2026 · ARXIV:2611.05748

THE PROBLEM

Reinforcement learning for large language models alternates between generation and training phases that have different hardware demands. The default GPU governor runs everything near peak clock frequency, causing memory-bound and waiting phases to consume excessive power without proportional throughput gains, while manual frequency tuning is too expensive and complex.

THE APPROACH. The authors introduce EE-RL (energy-efficient RL), an online controller that measures energy per valid training

WHY IT MATTERS

"By optimizing GPU clock frequencies dynamically during reinforcement learning post-training, this approach lowers electricity costs, reduces datacenter carbon footprints, and improves the sustainability of heavy AI training workloads without requiring tedious manual pre-tuning."

token (response tokens consumed by training that contribute to the loss) and throughput. It tests candidate frequency settings against the default GPU governor in short trials, selects the most energy-efficient setting within a throughput limit, and continuously adjusts frequencies as the workload changes.

KEY FINDINGS

- Fixed-frequency sweeps on H100 and GB200 show GPU energy savings of up to 28.2% per valid training token across synchronous and asynchronous execution while largely preserving throughput.
- EE-RL turns this hardware headroom into online savings, reducing GPU energy per valid training token by up to 16.9% at a modest throughput cost.
- Different RL phases (autoregressive generation vs. parallel training batch processing) expose distinct arithmetic intensities and hardware demands.
- Post-training an Olmo 3 32B model used 17x more datacenter energy than its instruction-tuned counterpart, highlighting the need for phase-aware energy optimization.

LIMITATIONS. Not stated in the provided text.

Filed under: reinforcement learning · gpu energy efficiency · dynamic voltage and frequency scaling

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ALSO ONLINE GitHub - yingchengyang/Reinforcement-Learning-Papers: Relate (github.com) · GitHub - Large-scale-Sustainable-Computing-LSC/green-code · (github.com) · Paper claims RL for reasoning only changes 1-3% of tokens, a (reddit.com)

REPORT · CS.CL · COMPUTATION AND LANGUAGE

Tug of Watts: A study on the Energy - Accuracy Trade-off of Large Language Models on Medical Question Answering Tasks

The paper investigates the energy-accuracy trade-offs of deploying Large Language Models (LLMs) for medical question answering tasks, demonstrating how to evaluate models using joint metrics rather than disjoint ones.

BY VIJU SUDHI, GIULIO BILLI, ANDREEA SCROB AND OTHERS · SUBMITTED 2 NOVEMBER 2026 · ARXIV:2611.02889

THE PROBLEM

While recent machine learning research focuses on incrementally improving downstream task accuracy in healthcare applications like medical question answering (QA), the associated energy consumption and carbon footprints during inference are frequently overlooked.

THE APPROACH. The authors examine 13 LLMs from 3 model families ranging from 270M to 14B parameters on medical QA tasks from the OpenMedical-LLM leaderboard. They use the CodeCarbon package to measure inference energy consumption and carbon emissions, and evaluate the models using joint trade-off metrics, Pareto-optimal frontiers, and a distance-based rating scheme.

KEY FINDINGS

- Investigated 13 LLMs from 3 model families ranging in size from 270M to 14B parameters.
- Larger models achieve higher task performance but at the cost of significantly higher relative carbon footprints.
- Reporting disjoint energy and accuracy metrics alone is insufficient for careful trade-off assessments; systematic joint metrics are required.
- Utilized Pareto-optimal frontiers and a distance-based rating scheme to clearly distinguish the trade-offs of individual models.

LIMITATIONS. Not stated in the provided text.

REPORT · CS.LG · MACHINE LEARNING

On-device Detection of Wearer-induced NLOS on Chest-worn UWB Tags: Machine Learning, Link Indicators, or Both?

The paper investigates the trade-offs between machine learning and lightweight link indicators for detecting wearer-induced non-line-of-sight (WNLOS) conditions in chest-worn UWB tags.

BY MARKUS GALLACHER, VU ANH MINH LE, MATTEO TROBINGER AND OTHERS · SUBMITTED 30 OCTOBER 2026 · ARXIV:2610.41799

THE PROBLEM

Wearer-induced non-line-of-sight (WNLOS) conditions cause significant UWB ranging errors, but existing detection methods often struggle to balance classification accuracy with the strict energy and runtime constraints of on-device processing.

THE APPROACH. The authors analyzed 21 classifiers—including machine learning (ML) models, radio link indicators (LI), and novel hybrid approaches—using a dataset of 1.5 million samples across various configurations.

KEY FINDINGS

- Relative performance of classifiers depends heavily on channel and PHY settings.
- Some ML techniques provide only marginal accuracy gains while consuming up to 39.6x more energy than lightweight alternatives.
- Hybrid classifiers effectively exploit the complementarity between ML and link indicators.
- A recommender tool was introduced to help users identify optimal classifiers based on specific energy and accuracy constraints.

LIMITATIONS. Not stated in the provided text.

WHY IT MATTERS. It addresses the sustainability of on-device intelligence by demonstrating that complex machine learning models are not always the most efficient choice, providing a tool to

WHY IT MATTERS. It addresses the soaring energy demands of real-time AI inference (projected to reach 652 TWh by 2030) by providing practitioners with methods to jointly optimize energy consumption and accuracy, helping them choose more efficient models and build sustainable machine learning pipelines.

Filed under: energy-accuracy trade-off · medical qa · sustainable llms

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ALSO ONLINE Paper page - 70B-parameter large language models in Japanese (huggingface.co) · Paper page - PeFoMed: Parameter Efficient Fine-tuning on Mul (huggingface.co) · Paper page - Enhancing Healthcare through Large Language Mod (huggingface.co)

optimize energy consumption without sacrificing necessary performance.

Filed under: uwb · energy-efficiency · wnlos

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ALSO ONLINE disi-unibo-nlp (DISI-UniBoNLP) · GitHub (github.com) · nlos · GitHub Topics · GitHub (github.com) · GitHub - cliansang/Identification-of-NLOS-and-MP-in-UWB-usin (github.com)

IN BRIEF · MORE FROM CS.LG

MedPsy: State-of-the-Art Small Medical Language Models for Efficient Edge Deployment

Davide Vitabile, Alexandro Buffa and others · cs.CL · 1 November 2026

LanG 2.0: Energy-Efficient Agentic AI for Semi-Automated Security Operations at Scale

Anes Abdennebi, Nadjia Kara and others · cs.AI · 29 October 2026

HyVox: Rethinking Low-Power Voice Intelligence towards Neuromorphic-von Neumann Edge Computing

Snehasis Addy, Eduardo Calle Ortiz and others · cs.NE · 1 November 2026

BitLLM: Efficient Low-Bit, Conversion-Hiding LLM Decoding on Mobile GPUs

Hangyu Zheng, Zhihao Shu and others · cs.AR · 31 October 2026

Binary Sparse Codes: Where are we on Energy Efficient Representations?

Jose Miguel Ramos, Luis SaCouto and others · cs.LG · 30 October 2026

Superscalar Loop Processors

Luca Colagrande, Pascal Etterli and others · cs.AR · 1 November 2026

ELSEWHERE IN THE JOURNALS

Peer-reviewed and web sources from the past month, beyond arXiv.

IEEEXPLORE.IEEE.ORG

Hybrid Phishing Detection Framework Using Green AI: An Energy-Efficient and Explainable Browser-Based Approach

carbon footprint of the system. With the real datasets from PhishTank, URLHaus, OTX, and Alexa, the real-time experiment of our system can achieve the performance of high 97.2 %...

IEEEXPLORE.IEEE.ORG

Green-Aware Connectome Classification Using Functional Connectivity: A Cross-Device Power Consumption Analysis

The rapid growth of machine learning has raised concerns about the environmental impact of computational models. The emerging paradigm of Green AI emphasizes the importance of...

SCIENCEDIRECT.COM

Toward sustainable on-device intelligence: A survey on energy-efficient RAG systems with small language models

Pillar 3: Green AI and Sustainability. A growing body of research has documented the environmental cost of the AI scaling paradigm. Strubell et al. showed that training a single...

SCIENCEDIRECT.COM

Decarbonization dispatch for AI computing center parks considering task-level carbon traceability

Large-model training, batch inference, and multimodal services are rapidly increasing the energy, water, and carbon footprint of AI server infrastructure . De Vries-Gao notes that...

SECTION · ARXIV EESS.SY

Systems and Control

Also drawing on eess.SP, cs.LG, stat.ML, physics.app-ph, cond-mat.mtrl-sci. Coverage: battery management systems: state-of-charge and state-of-health estimation, degradation modelling and remaining useful life prediction.

THE WEEK IN BRIEF

3 new papers in eess.SY · Systems and Control made the cut this week. Highlights include: FlowBatt: Flow Matching for Probabilistic Battery Degradation Prediction; Frequency-Calibrated Decision-Focused Learning for BESS Dispatch Under Renewable Uncertainty.

LEAD STORY · CS.LG · MACHINE LEARNING

FlowBatt: Flow Matching for Probabilistic Battery Degradation Prediction

FlowBatt is a conditional generative framework that uses flow matching and a diffusion transformer to predict probabilistic battery state-of-health (SOH) trajectories.

BY HAMIDREZA EIVAZI, ESLAM SHARAAWY, STEFAN WITTEK AND OTHERS · SUBMITTED 2 NOVEMBER 2026 · ARXIV:2611.03150

THE PROBLEM

Existing machine learning methods for battery health primarily focus on deterministic point predictions for remaining useful life (RUL) or SOH, failing to capture the probabilistic nature of degradation trajectories across diverse chemistries.

THE APPROACH. The authors introduce FlowBatt, which uses a diffusion transformer (DiT) backbone to model full SOH degradation trajectories conditioned on early-cycle capacity data, while using explainable AI to optimize the input encoder.

KEY FINDINGS

- Achieved the best trajectory-wise SOH prediction performance across five public benchmark datasets.
- Maintained prediction errors below 2.5% for SOH trajectories.
- Delivered competitive RUL prediction performance compared to specialized baselines.
- Provided empirical prediction intervals by generating multiple plausible degradation trajectories.
- Reduced random forest RMSE and MAPE on the MATR-1 dataset by approximately 26% and 19% respectively using synthetic trajectories.

WHY IT MATTERS

"It shifts battery management from simple point-based RUL estimation to probabilistic trajectory modeling, allowing for better uncertainty quantification and more robust health forecasting across heterogeneous battery chemistries."

LIMITATIONS. The model lacks direct inputs for environmental factors like temperature or charging protocols, does not enforce physical constraints, and shows higher RUL errors and undercoverage of prediction intervals on specific datasets (MATR-2 and HUST).

Filed under: flow matching · battery degradation · probabilistic forecasting

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ALSO ONLINE GitHub - NY1024/AgentSafety-Papers: Daily Tracking of LLM Ag (github.com) · 18839_Generative_Modeling_of_B.pdf (openreview.net) · GitHub - AkashDas-AI/Battery-Degradation-Prediction-for-Ener (github.com)

REPORT · EESS.SY · SYSTEMS AND CONTROL

Frequency-Calibrated Decision-Focused Learning for BESS Dispatch Under Renewable Uncertainty

Battery energy storage system (BESS) flexibility depends on state of charge and intertemporal operating constraints.

BY ZIHAN MENG, CHENXI HU, XIN CUI AND OTHERS · SUBMITTED 30 OCTOBER 2026 · ARXIV:2610.41238

THE PROBLEM. Not stated in the provided text.

THE APPROACH. Renewable forecast errors of similar magnitudes can lead to different BESS dispatch results. An error may alter early charging or discharging decisions and reduce the flexibility available for subsequent regulation, while another may have little effect on the resulting schedule.

KEY FINDINGS

- To account for these unequal consequences, this paper proposes Frequency-Calibrated Decision-Focused Learning (FC-DFL), which trains renewable predictors using the realized operating costs of their induced BESS schedules.
- Gaussian smoothing and solver-based zeroth-order estimation provide training feedback without requiring differentiating through the non-differentiable mixed-integer BESS dispatch problem.

REPORT · CS.LG · MACHINE LEARNING

Cross-Domain Joint-Embedding Pretraining For Heterogeneous Industrial Time Series

Industrial predictive maintenance and manufacturing tasks frequently contain abundant sensor measurements, but with limited labeled data.

BY JONATHAN ADAM RICO, NAGARAJAN RAGHAVAN AND SENTHILNATH JAYAVELU · SUBMITTED 2 NOVEMBER 2026 · ARXIV:2611.02489

THE PROBLEM. Not stated in the provided text.

THE APPROACH. We investigate cross-domain self-supervised pretraining using joint-embedding predictive architecture for heterogeneous industrial time series (JEPA-HITS), which learns one encoder from various domains such as battery, semiconductor chemical mechanical planarization (CMP), computer numerical control (CNC) machining, and bearing while preserving dataset-specific temporal scales. We evaluate representation reuse on regression and classification problems under reduced-label downstream training, and leave-one-domain-out (LODO) pretraining.

KEY FINDINGS

- Compared with the same architecture initialized randomly, LODO pretraining reduces RMSE from 10.72 to 7.95 on battery remaining useful life (RUL), 4.34 to 3.89 on CMP, and 13.53 to

- The same decision feedback is further combined with prediction sensitivity to develop a decision-oriented XAI method that identifies the input features and forecast periods most consequential to BESS operation.

LIMITATIONS. Not stated in the provided text.

WHY IT MATTERS. Matched the eess.SY topic filter: battery management systems: state-of-charge and state-of-health estimation, degradation modelling and remaining useful life prediction.

Filed under: state of charge · estimation

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ALSO ONLINE Daily Papers - Hugging Face (huggingface.co) · 217 kB - Hugging Face (huggingface.co) · vocab.txt · allenai/aspire-sentence-embedder at 340520d7e4d7 (huggingface.co)

Editor's note: this report was prepared from the abstract alone.

11.72 on tool wear regression, while improving bearing fault diagnosis macro-F1 from 0.547 to 0.737.

- Overall results suggest that unlabeled time series measurements from heterogeneous industrial processes can provide reusable representations for downstream learning with limited labeled data.

LIMITATIONS. Not stated in the provided text.

WHY IT MATTERS. Matched the eess.SY topic filter: battery management systems: state-of-charge and state-of-health estimation, degradation modelling and remaining useful life prediction.

Filed under: remaining useful life

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ALSO ONLINE Learning to Factorize Spatio-Temporal Foundation Models (openreview.net) · GitHub - AI-in-Transportation-Lab/awesome-jepa: A carefully (github.com) · GRAVER: Generative Graph Vocabularies for Robust ... (openreview.net)

Editor's note: this report was prepared from the abstract alone.

ELSEWHERE IN THE JOURNALS

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SCIENCEDIRECT.COM

Domain adaptation state of health estimation for lithium-ion batteries accounting for state of charge dependence

Data-driven methods regard SOH estimation as a black-box problem. Through extracting aging-related features from operational data and training machine learning models with...

SCIENCEDIRECT.COM

Physics-informed neural networks for battery modeling and state estimation: A methodological review

constraints and the corresponding strategies for embedding such knowledge into learning frameworks. It further summarizes the major applications of PINNs in battery management,...

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Embedded artificial intelligence in battery management systems: Pruning and quantization for efficient state-of-charge and state-of-health estimation

IEEE Access (2020) Y. Gao _et al._ ### Co-estimation of state-of-charge and state-of- health for lithium-ion batteries using an enhanced electrochemical model ### IEEE Trans....

SCIENCEDIRECT.COM

Tokenizing battery degradation: a physics-informed and uncertainty-aware framework for state-of-health estimation

To overcome the limitations of physics-based approaches, the field has increasingly turned to data-driven methods that leverage machine learning to learn degradation patterns...

COLOPHON

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